k-vector angular correlations in negative refraction for TM polarization in nanosphere dispersed liquid crystal (NDLC) metamaterial

G. Pawlik[a], W. Walasik[b,c], K. Tarnowski[a], A.C. Mitus[a], I. C. Khoo[d]

[a]Institute of Physics, Wroclaw University of Technology, Wybrzeze Wyspianskiego 27, Poland
[b]Aix Marseille Universit, CNRS, Centrale Marseille, Institut Fresnel, UMR 7249, 13013 Marseille, France
[c]Institut de Ciències Fotòniques, Universitat Politècnica de Catalunya, Castelldefels, Barcelona 08860, Spain
[d]Department of Electrical Engineering, Pennsylvania State University, University Park, PA, 16802 USA

ABSTRACT

We study the behavior of refracted angle for k-vector at the interface of uniaxial anisotropic media in the case of nanosphere dispersed liquid crystal (NDLC) matematerial. Finite Element (FE) calculations (COMSOL Multiphysics) are used to trace the propagation of the electromagnetic wave. Preliminary results on the influence of incident angle on refracted angle wave-vector are presented.

1. INTRODUCTION

The material consisting of coated spheres dispersed in liquid crystal host was proposed as a medium with a negative refractive index. The possibility of tuning of this metamaterial was discussed by Khoo and co–workers[1] and was investigated using Monte Carlo method and Finite Element (Comsol Multiphysics) calculations.[2–5]

Recently we have shown[6] that the NDLC metamaterial allows to create negative refraction for the Poynting vector for a wide range od incident angles. This effect was theoretically analyzed for TM polarized wave at infrared frequencies. The objective of this paper is the calculation and analysis of refracted angle for k-vector because for considered anisotropic media the angle between k-vector and S-vector is non-zero.

2. REFRACTED ANGLE FOR K-VECTOR IN NDLC FOR PLANAR ALIGNMENT OF NLC HOST

We have studied a rectangular NLC cell with the thickness L, filled with NDLC metamaterial (Fig. 1 (a)). The parameters of coated nanospheres (Fig. 1 (b)) are given in Table 1 and are the same as in Ref.[6] In this paper we analyze a simple case only when the molecules of NLC are parallel to each other and are parallel to the cell boundaries (planar alignment). Ordinary and extraordinary indices of refraction of NLC molecules were $n_o = 1.41$ and $n_e = 1.7$ (for details see Ref.[6]).

Angle of refraction for the Poynting vector $\theta_{r,S}$ was calculated from pictures with the magnetic field mapping presented in Ref.[6] Dependence of this angle on incident angle is presented in Fig. 2.

Behavior of the k-vector of refracted wave was studied using the gaussian wave beam with the wavelength $\lambda_0 = 1.74\,\mu$m. The wave was polarized in $x - y$ plane with magnetic field along z-axis (TM polarization). Effective values of electromagnetic tensor components were calculated using equations from Ref.[6]

The results of FE calculations of the distribution of magnetic field of incoming and outgoing electromagnetic waves are shown in Fig. 3. We have here two examples with incident angles $\theta_i = 30°$ (a) and $\theta_i = 50°$ (b). In each case the refracted angle for the wave-vector is positive. Direction of k-vector is represented by arrow, which is normal to the surfaces of constant phase (wave fronts).

Send correspondence to grzegorz.pawlik@pwr.wroc.pl

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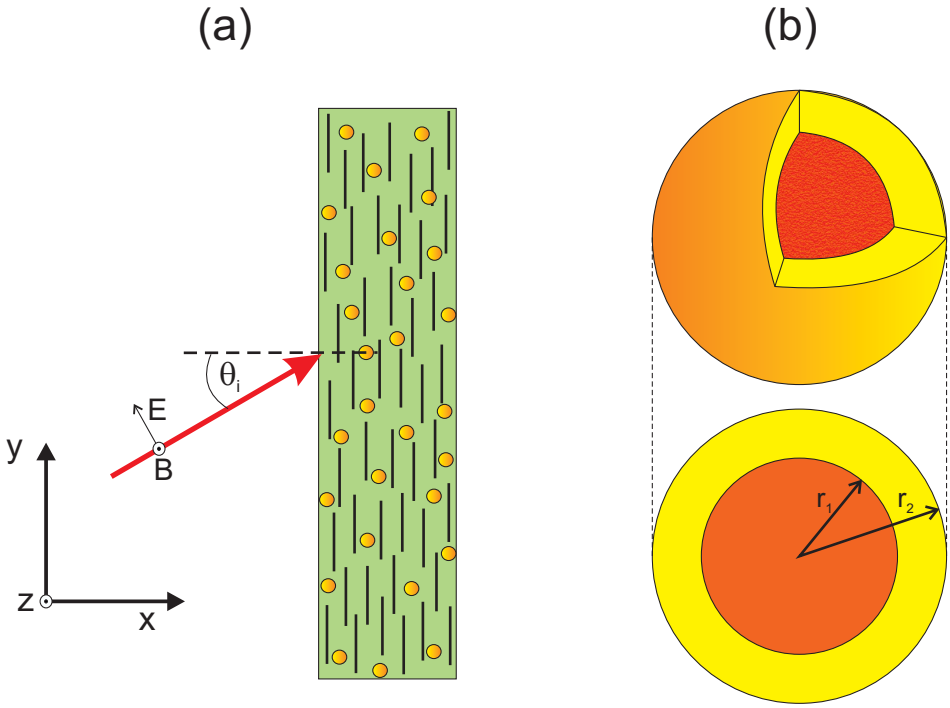

Figure 1. Geometry of the studied system: cell filled with NDLC metamaterial and incident wave (a) and cross section of a coated nanosphere (b).

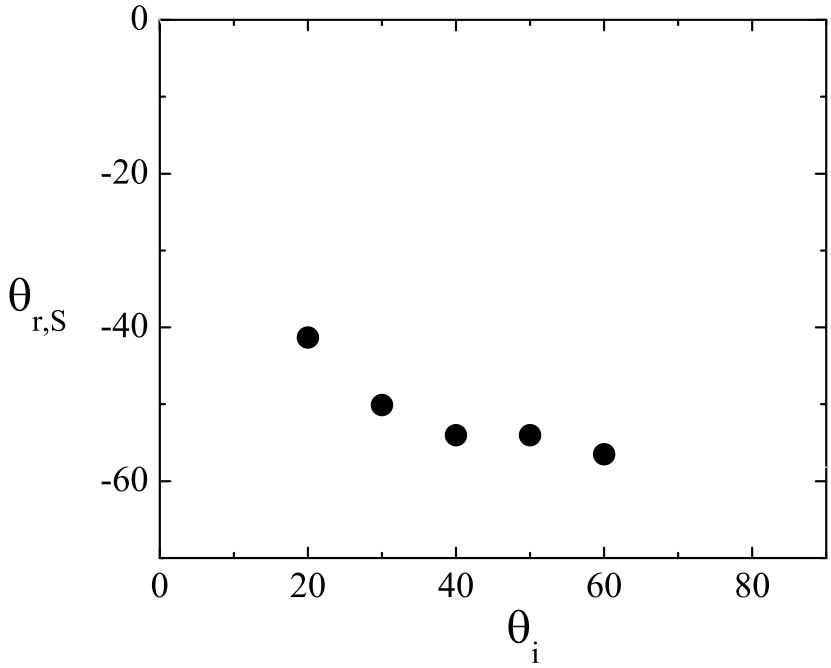

Figure 2. Dependence of refraction angle for the Poynting vector on incident angle (angles in degrees).

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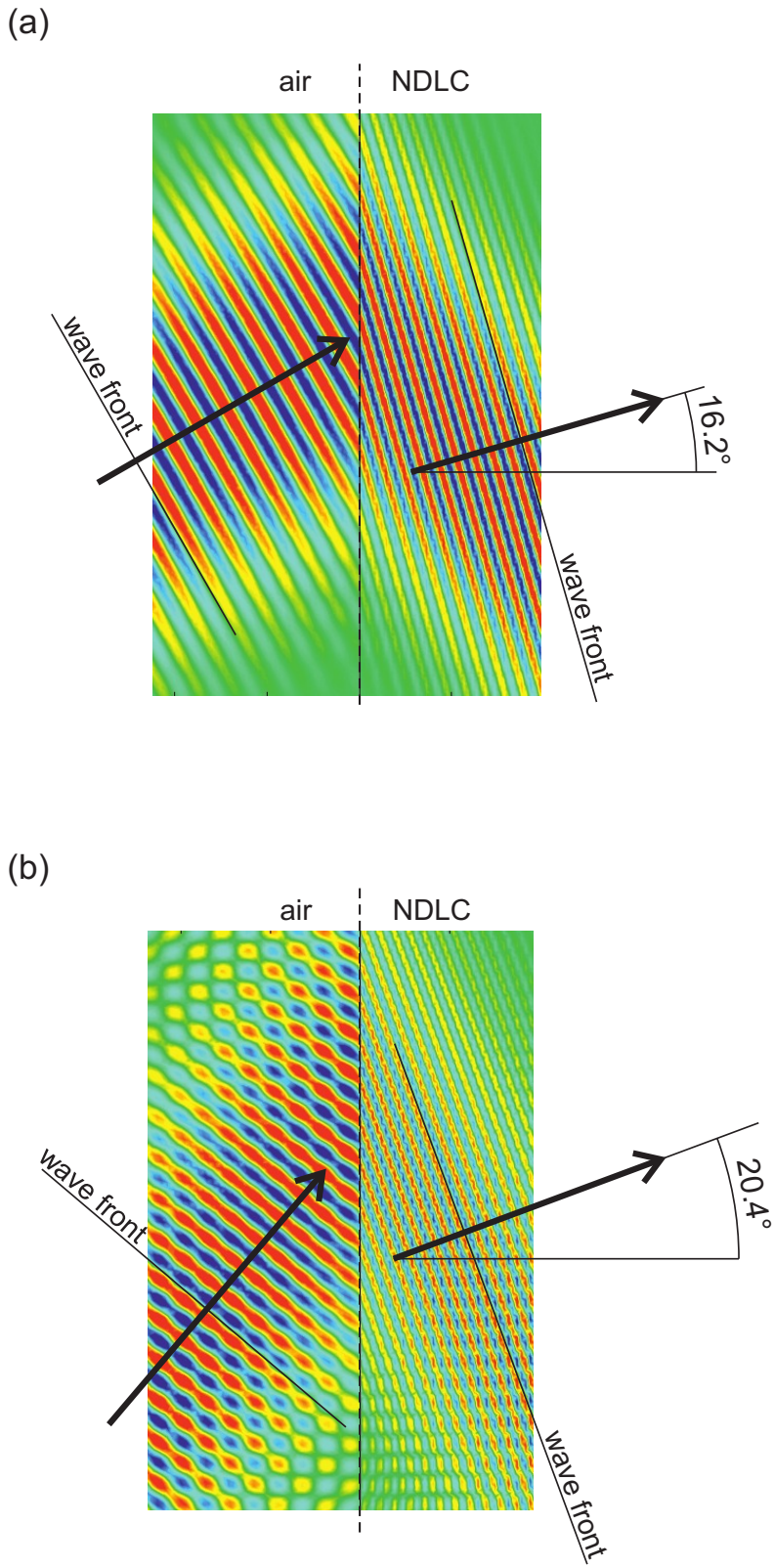

Figure 3. Finite Element calculation of the magnetic field mapping for gaussian beam with TM illumination: $\theta_i = 30^\circ$ (a) and $\theta_i = 50^\circ$ (b). Arrows indicate direction of k-vectors.

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r_1	0.05μm
r_2	0.06μm
f	0.06
$\varepsilon(\infty)$	17
$\omega_T/2\pi$	360THz
$\omega_L/2\pi$	855THz
$\gamma_1/2\pi$	3.75THz
$\omega_p\pi$	230THz
γ_2	$\omega_p/60$

Table 1. Parameters used in nanosphere design. f is the filling factor, $\epsilon_1(\infty)$ denotes the high–frequency dielectric constant, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively, ω_p denotes plasma frequency and γ_2, γ_1 – damping coefficients.

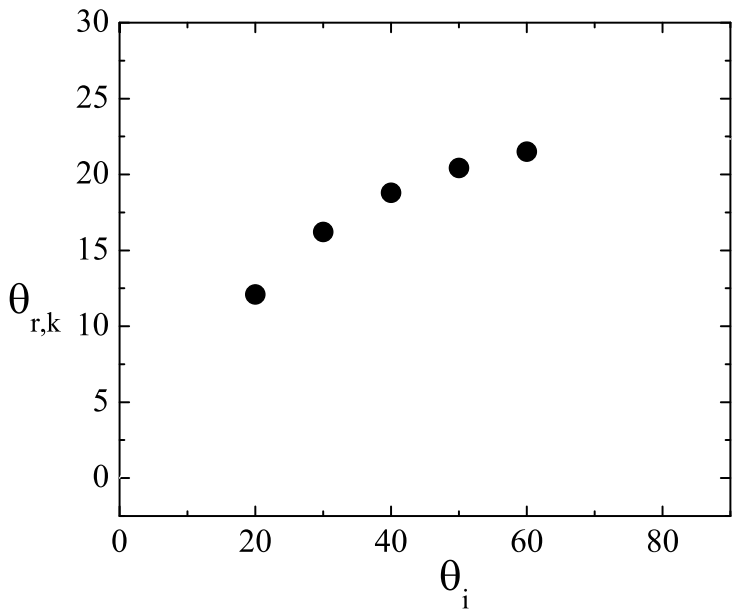

Figure 4. Dependence of refraction angle for k-vector on incident angle(angles in degrees).

3. SUMMARY

We have presented preliminary results related to the influence of incident angle on refracted angle for k-vector at the interface of uniaxial anisotropic media for case with negative refraction for TM wave. To this end, we have used Finite Element simulations (COMSOL Multiphysics) of the nanosphere dispersed liquid crystal metamaterial to calculate the angles of refraction. We have shown that when negative refraction is present, the refracted angle for k-vector can be positive for considered anisotropic material.

Acknowledgements

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REFERENCES

1. Khoo, I. C., Werner, D. H., Liang, X., Diaz, A. and B. Weiner, "Nanosphere dispersed liquid crystals for tunable negativezeropositive index of refraction in the optical and terahertz regimes", Opt. Lett. 31, 2592 (2006).

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2. Pawlik, G., Jarema, M., Walasik, W., Mitus, A.C, and Khoo, I.C., "Field-induced inhomogeneous index distribution of a nano-dispersed nematic liquid crystal metamaterial near the Freedericksz transition: Monte Carlo studies", J. Opt. Soc. Am. B 27, 567 (2010).
3. Pawlik, G., Walasik, W., Mitus, A.C., and Khoo, I.C., "Large gradients of refractive index in nanosphere dispersed liquid crystal metamaterial with inhomogeneous anchoring: Monte Carlo study", Opt. Mater. 33, 1459 (2011).
4. Pawlik, G., Walasik, W., Jarema, M., Mitus, A.C., Khoo, I.C., "Temperature dependence of gradient of refractive index in nanosphere dispersed liquid crystal (NDLC) metamaterial at infrared frequencies: Monte Carlo study", Opt. Mater. 34, 1656 (2012).
5. Pawlik, G. Tarnowski, K., Walasik, W. Mitus, A.C., Khoo, I.C., "Infrared cylindrical cloak in nanosphere dispersed liquid crystal metamaterial", Opt. Lett. 37, 18471849 (2012).
6. Pawlik, G., Walasik, W., Tarnowski, K., Mitus, A.C., and Khoo, I. C., "Negative refraction for TM polarization in nanosphere dispersed liquid crystal (NDLC) metamaterial", Proc. SPIE 8828, 88280E (2013). doi:10.1117/12.2025176.

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